SECTIONS ON THE SPECTRUM OF A STRICTLY HENSELIAN LOCAL RING ARE THE STALK AT THE CLOSED POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three assertions of the statement.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Strictly henselian local ring and its etale site. Same chapter, the description of the etale site of the spectrum of a strictly henselian local ring, in which every etale sheaf is determined by its stalk at the closed point. Supplies the identification of global sections with the stalk.
- Stalk at the closed point and exactness of the stalk functor. Same chapter, the computation of stalks at the closed point and the exactness of taking stalks. Supplies the exactness of global sections and the vanishing of higher cohomology.

Source reference. Original theorem 03QH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4157–4157. Navigation pointer invoked at proof line 8226; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 015F: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8241–8241. Navigation pointer invoked at proof line 8241; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a strictly henselian local ring. Set $S = \mathrm{Spec}(R)$ and let $\overline{s}$ be its closed point. Then the global sections functor $\Gamma(S, -) : Ab(S_{étale}) \to Ab$ is exact. In fact we have $\Gamma(S, \mathcal{F}) = \mathcal{F}_{\overline{s}}$ for any sheaf of sets $\mathcal{F}$. In particular*

$$\forall p \geq 1, \quad H^p_{étale}(S, \mathcal{F}) = 0$$

for all $\mathcal{F} \in Ab(S_{étale})$.

Proof. If we show that $\Gamma(S, \mathcal{F}) = \mathcal{F}_{\overline{s}}$ then $\Gamma(S, -)$ is exact as the stalk functor is exact. Let $(U, \overline{u})$ be an étale neighbourhood of $\overline{s}$. Pick an affine open neighborhood $\mathrm{Spec}(A)$ of $\overline{u}$ in U. Then $R \to A$ is étale and $\kappa(\overline{s}) = \kappa(\overline{u})$. By Theorem 03QH we see that $A \cong R \times A'$ as an R-algebra compatible with maps to $\kappa(\overline{s}) = \kappa(\overline{u})$. Hence we get a section

$$\begin{array}{ccc} \mathrm{Spec}(A) & \longrightarrow & U \\ & \nwarrow & \downarrow \\ & & S \end{array}$$

It follows that in the system of étale neighbourhoods of $\overline{s}$ the identity map $(S, \overline{s}) \to (S, \overline{s})$ is cofinal. Hence $\Gamma(S, \mathcal{F}) = \mathcal{F}_{\overline{s}}$. The final statement of the lemma follows as the higher derived functors of an exact functor are zero, see Derived Categories, Lemma 015F. □